

A city, explicit time, and actual observations

CREBAIN native environment | Source component guide | September 5, 2026

CREBAIN advances its existing Rapier drone dynamics. One authored metric scene supplies building colliders, meshes, Gaussian surfaces, acoustic obstruction, and thermal visibility.

The CPU owns physical state, controller memory, rotor response, acoustic history, and temperature. A private Node and Chromium process owns actual RGB and thermal pixel rendering.

What one step means

$$t_k = k / 120 \text{ seconds}; \quad \Delta t = 1 / 120 \text{ seconds}$$

The integer k is the completed physics tick. The next scheduled action applies before that tick. Display reads and wall-clock delays cannot advance simulation time.

Three ticks span 25 milliseconds. Six ticks span 50 milliseconds. These are exact interface durations; the integrator uses the existing floating-point time step.

Command and response are different

A motor command is a named fraction from 0 to 1. An attitude command sets roll and pitch in radians, yaw rate in radians per second, and altitude in meters. The controller retains its clamps, mixer, and saturation.

The frame is positive Y upward, positive Z forward, and positive X right. It is not ENU or PushT. Commanded acceleration must never be claimed as delivered acceleration without a separate measured adapter.

The default attitude controller failed a frozen three-second tracking campaign. A separate twelve-tick free-rotation probe observed no Euler gyroscopic evolution in its inspected configuration. The one-second city export does not qualify tracking or physical fidelity.

Output and reference stay separate

Accepted batches bind actual pressure, RGB, and thermal arrays. Privileged CPU truth is a separate field. A predictor must not gain checkpoint or future-label access through an observation interface.

This component is not physical calibration, installed desktop qualification, a real-vehicle controller, or a completed NCP or Prisoma experiment integration.

Heat: from rotor power to a thermal pixel

Declared lumped gray-surface model | Units and worked examples

For rotor r , torque τ_r is measured in newton-meters. Angular speed ω_r is measured in radians per second. Actual rotor state supplies mechanical power P_m in watts.

$$P_m = \sum_r |\tau_r \omega_r|$$
$$P_e = P_m / \eta \quad P_h = P_m (1/\eta - 1)$$

Motor efficiency η is dimensionless, with $0 < \eta \leq 1$. Electrical input P_e and heat input P_h are declared constant-efficiency quantities. The normalized battery value is not their energy source.

Example: 70 W of mechanical power at efficiency 0.7 requires 100 W of modeled electrical input and gives 30 W of modeled heat.

Temperature T is in kelvin. Ambient temperature is T_a . Heat capacity C is in J/K, effective area A in m^2 , and convection coefficient h in $W/(m^2 K)$. Emissivity ε is dimensionless.

$$C (T_{k+1} - T_k) / \Delta t = P_h$$
$$- h A (T_{k+1} - T_a)$$
$$- \varepsilon \sigma A (T_{k+1}^4 - T_a^4)$$

The constant σ is $5.670374419 \times 10^{-8} W/(m^2 K^4)$. The next temperature appears in the loss terms: this is backward Euler.

The residual derivative $C/\Delta t + hA + 4\varepsilon\sigma AT^3$ is positive. A bounded bracket and 64 bisection iterations therefore select one admitted solution. An out-of-range solution fails the transition.

Example: With no heat loss, 30 W and $C = 100$ J/K increase temperature by 0.0025 K in one tick. Positive losses reduce this increase above ambient.

Radiance is not temperature

Actual floating-point rendering | No spectral camera calibration

$$L = (\sigma / \pi) [\epsilon T^4 + (1 - \epsilon) T_a^4]$$

The output L is diffuse bolometric radiance in $W/(m^2 \text{ sr})$. A steradian, sr, measures solid angle. The reflected term assumes uniform ambient radiation.

Ambient example: When $T = T_a = 293.15 \text{ K}$, L is about $133.2973 \text{ W}/(m^2 \text{ sr})$, regardless of emissivity.

Warm-surface example: $T = 400 \text{ K}$, $\epsilon = 0.9$, and the same ambient temperature give about $429.1870 \text{ W}/(m^2 \text{ sr})$. Independent arithmetic checks verify these actual rendered values.

What the image contains

An actual float render target resolves visible surfaces against the shared scene geometry. Each thermal pixel stores one float32 little-endian radiance value. An opaque wall can hide a warm drone.

The RGB path separately awaits actual Spark and Three.js rendering. It copies RGBA8 sRGB bytes before another capture can reuse the buffer. Rows begin at the bottom-left.

What this model does not contain

Bolometric means all wavelengths. This is not an 8-14 micrometer camera, a detector spectral response, or a temperature estimate. No false-color RGB output is presented as a physical measurement.

Thermal area, heat capacity, emissivity, and efficiency are declared effective parameters. They are not inferred from the visible drone mesh. The model does not claim measured drone fidelity or a globally closed scene energy balance.

A decisive image control

Setting Gaussian opacity to zero versus one must change actual RGB pixels. Mesh rendering alone cannot satisfy the Gaussian contribution claim. Camera order, cold contexts, and repeated moving-drone captures are separate controls.

Pressure: phase, delay, and an honest range limit

Explicit acoustic forward model | 16,000 samples per second

$$[\text{floor}((k - 1) f_s / 120), \text{floor}(k f_s / 120))$$
$$f_s = 16,000 \text{ Hz}$$

The first three complete intervals contain 133, 133, and 134 samples. CPU advances first; completed pose and rotor speed remain constant within that audio block. This is a declared approximation.

$$f_b = 2 n / 60 \text{ Hz}$$
$$q(\varphi) = \sin \varphi + 0.3 \sin(2\varphi) + 0.1 \sin(3\varphi)$$

For a two-bladed rotor, n is revolutions per minute and f_b is blade-passage frequency. At 6,000 RPM the fundamental is 200 Hz. Phase φ advances at that frequency. Amplitude scales with $(n/15,000)^2$ and the declared reference pressure.

$$\text{delay} = \max(d, d_0) / c \text{ seconds}$$
$$\text{gain} = d_0 / \max(d, d_0) \times \text{obstruction factor}$$

Distance d and reference distance d_0 are in meters. Sound speed c is in m/s. Fractional delays interpolate retained waveform samples. At $d = 34.3 \text{ m}$ and $c = 343 \text{ m/s}$, delay is 0.1 seconds, or 1,600 samples.

That example needs a maximum range of at least 34.3 m. The default 32 m profile omits the source. Noise still produces valid microphone samples; valid output does not imply visibility of every intervention.

The selected 256-drone challenge changed dynamics, RGB, and thermal output. Its selected drone stayed more than 44 m from either microphone, so the intervention did not change pressure. The failed all-modality assertion is retained.

The model retains phases, delay rings, noise generator state, and sample index. It omits echoes, diffraction, calibrated directivity, and moving-source retarded geometry. Cloned noise is shared randomness, not independent replication.

A checkpoint is a bounded causal contract

Exact CPU reconstruction | Fresh static renderer | Independent branch lifetime

A live owner-issued checkpoint requires a released observation lease, no pending operation, and a tick at which every camera was due. If camera periods have no common positive tick within 7,200 ticks, that checkpoint capability is unavailable.

The complete CPU checkpoint includes dynamics, controller memory, motors, battery, accepted future actions, acoustic history, noise, and temperature. The same actual transition reconstructs it from the bound initial state and action prefix.

$$S_k = F(S_{k-1}, u_k)$$

S is complete admitted CPU state. F is the existing transition. u contains the selected controls for that tick, including no replacement. Exact canonical state comparison guards replay acceptance.

A child starts a fresh private static renderer and must reproduce the checkpoint's current typed pixel digests. This does not clone hidden GPU state. Equal current images cannot prove all future images equal.

Both experimental arms use fresh matched siblings. A warm parent is an additional diagnostic, not the sole causal control. Frozen same-action future outputs must remain exactly equal. Deliberately different action effects are allowed.

Each child binds its parent checkpoint, accepted batch prefix, and action-history position. New owner and graphics generation identities make causal batch hashes different even when raw samples match. Audit JSON alone cannot recreate a live fork token.

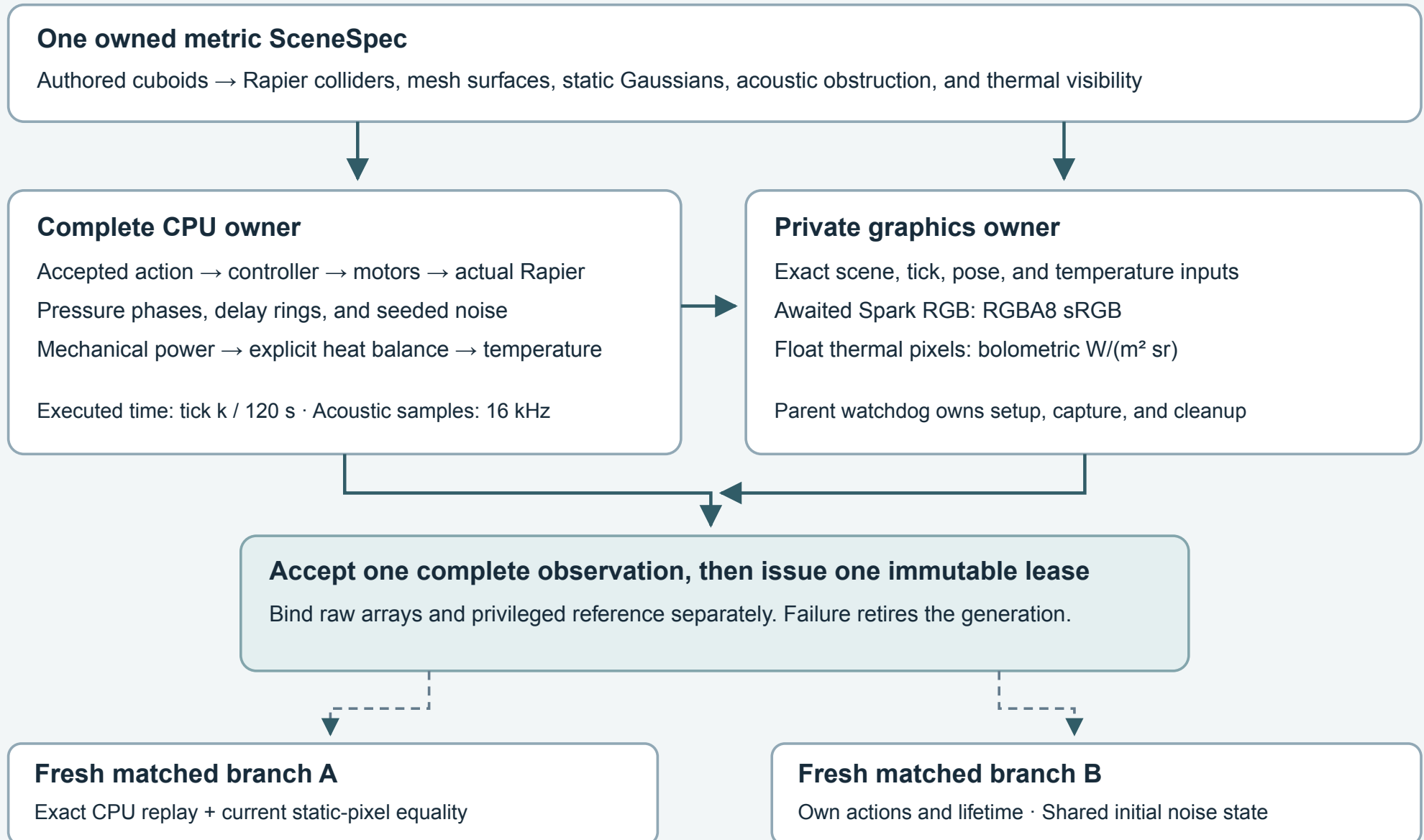
Failure and capacity

Failed reconstruction retires only the candidate. Primary failure and cleanup failure remain separate. Constructing and unresolved generations still consume the four-owner family ceiling. A no-op retirement retry cannot erase an earlier unresolved CPU cleanup.

The 128 MiB retained CPU budget is separate from one 64 MiB temporary reconstruction checkpoint. Camera staging reserves at most 32 MiB raw bytes. A CPU restore also reserves one extra temporary owner. Unresolved cleanup retains that slot and the 64 MiB checkpoint reservation, blocking another reconstruction. These are logical reservations, not total RSS or GPU-memory guarantees.

CREBAIN · A city, explicit time, actual observations

Standalone source component · +Y up, +Z forward · No neural controller or protocol bundle required



Dashed links: checkpoint reconstruction at an all-camera barrier. Hidden GPU-state cloning and universal future equality are not claimed.